

Text Image GenAI methods summary

STUDY PACK

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Text-Image-GenAI — Methods (1-page summary)

Purpose

- Build a modular pipeline that performs text→image generation and image→text captioning, enabling round-trip evaluation and iterative improvement.

Pipeline Overview

- Components: (1) Text encoder, (2) Latent diffusion model (text→image), (3) Vision-language encoder/captioner (image→text), (4) Evaluation & API layer.
- Data flow: Prompt → text embedding → diffusion sampling → generated image → vision encoder → caption → compare to prompt.

Modeling Details

- Latent diffusion: operates in a compressed latent space to iteratively denoise a noise tensor conditioned on text embeddings. Advantages: high fidelity, controllable via classifier-free guidance.
- Vision-language encoders: BLIP/BLIP-2 or similar transformer-based models map images and text into a shared embedding space for captioning and semantic comparison.
- Conditioning & control: use guidance scale and temperature parameters; classifier-free guidance mixes conditional/unconditional denoising to trade fidelity vs diversity.

Training & Objectives

- Multi-objective loss: $L(\Theta) = \alpha \cdot L_{\text{text}} + \beta \cdot L_{\text{img}} + \gamma \cdot L_{\text{perc}} + \lambda \cdot R(\Theta)$.
- L_{text} : CLIP-style contrastive loss to align text↔image embeddings.
- L_{img} : diffusion reconstruction/denoising loss (mean squared error in latent/noise prediction).
- L_{perc} : perceptual loss (VGG features) to preserve visual quality.
- $R(\Theta)$: regularization (weight decay, priors).

Evaluation & Round-Trip

- Roundtrip: generate image from prompt, caption image, then compute semantic alignment between original prompt and generated caption.
- Metrics: FID (image quality), CLIPScore (semantic alignment), BLEU/METEOR/ROUGE/CIDEr (caption quality).
- Ablations: vary model sizes, guidance scales, and captioner architectures to measure tradeoffs.

Engineering & Deployment

- Modular design: decouple diffusion model, captioner, and API service (FastAPI). Allows swapping components and offline vs cloud runtimes.
- Efficiency: use latent-space sampling, mixed precision and caching of embeddings; batch captioning during evaluation.
- Scalability: expose endpoints for text→image and captioning; worker queue for heavy generation jobs.

Practical Notes

- Compute: diffusion sampling is CPU/GPU heavy; consider smaller checkpoints or low-rank adapters for constrained environments.
- Ethics & Safety (brief): add watermarking, content filters, and human review for sensitive outputs.

References (key models)

- Latent diffusion / Stable Diffusion family; BLIP / BLIP-2 for captioning; CLIP for embedding alignment.

Generated as a concise, one-page methods summary for exam prep and quick reference.